

# Spherical S/D/R and Four State Patterns

Spherical geometry

Hierarchical decision

$$d_g(a, b) = \arccos(\text{clip}(a^T b, -1, 1))$$

$S > \kappa$ ? Confusion

$$\mu_c = \text{norm}(\sum_p z_{c,p})$$

$D \leq \tau$ ? Consensus

$$s_c = P^{-1} \sum_p d_g^2(z_{c,p}, \mu_c); S = (s_1 + s_2 + s_{12})/3$$

$|R| \leq \delta_i$ ? Balanced

$$D = d_g(\mu_1, \mu_2) / (\sqrt{s_1 + s_2} + \varepsilon)$$

$$R = [d_g(\mu_{12}, \mu_2) - d_g(\mu_{12}, \mu_1)] / [d_g(\mu_1, \mu_2) + \varepsilon]$$

otherwise Dominant

$R > 0$ : V lean;  $R < 0$ : T/A lean